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MILITARY SPECIFICATION

FLOODLIGHT SET, ELECTRIC, DIESEL-ENGINE-DRIVEN, TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a self-contained, trailer-mounted, floodlight set, equipped with a diesel-engine-driven, 6 kilowatt, generator set and either a mechanical or pneumatically operated telescoping floodlight tower.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596	- Connector, Electrical, Power, General Specification for.
W-C-596/12	- Connector, Receptacle, Electrical, General Purpose, Duplex, Hospital Grade, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts. 50/60 Hertz.
W-C-596/99	- Connector, Receptacle, Electrical, Special Purpose, Duplex, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.
W-C-596/100	- Connectors, Plug, Electrical, Special Purpose, Grounding, 2 Pole, 3 Wire, 15 Amperes, 125 Volts, 50/60 Hertz.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6230

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QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-C-104 - Preservation, Methods of.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture-and Fungus-Resistant, for Treatment of Communications, Electronic, and Associated Equipment).
MIL-T-704 - Treatment and Painting of Materiel.
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000 Pound Capacity.
MIL-A-8421 - Air Transportability Requirements.
MIL-G-28554 - Generator Set, Mobile-Electric-Power and Supplemental Equipment, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers And Trailer Dollies; Preservation and Packaging of.
MIL-STD-461 - Electromagnetic Interference.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1223 - Administrative Wheeled Vehicles, Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.
MS51336 - Lunette-Coupler, Drawbar, Ring.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense (DoD) adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

AB 1 - Molded Case Circuit Breakers.
WD 1 - General-Purpose Wiring Devices.
WD 5 - Specific-Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 62 - Standard for Flexible Cord and Fixture Wire.
UL 486 - Wire Connectors and Soldering Lugs for Use With Copper Conductors.
UL 489 - Molded-Case Circuit Breakers and Circuit-Breaker Enclosures.
UL 498 - Electrical Attachment Plugs and Receptacles.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The floodlight set shall consist, basically, of a trailer, mounted with a diesel-engine-driven generator, telescoping tower, portable masts, and eight flood lights (four 1,000 and four 500 watt (W) units). The floodlights and all electrical equipment shall be provided with a weatherproof storage compartment.

3.1.1 Dimensions and weight. The overall dimensions of the floodlight set, with attachments retracted and stowed in towable condition, shall not exceed 15 feet in length, 7 feet in width, 6 feet in height, or 2,500 pounds (lb) in weight.

3.1.2 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3.2 First article. The manufacturer shall furnish one floodlight trailer for first article inspection, tests and approval (see 4.3 and 6.2.1).

3.3 Standard product. Except where modified herein, it is intended that the equipment and its component parts shall be a regular commercial product of the manufacturer or his suppliers. The parts shall be new and unused.

3.3.1 Maintenance design. The design of the equipment shall be in accordance with current engineering practices. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions that may be hazardous to personnel or deleterious to equipment.

3.3.2 Screw threads. All bolts, nuts, and screws shall have standard screw threads in accordance with FED-STD-H28.

3.4 Alternator. The alternating current (ac), alternator shall be a brushless design with a rated output at between 1600 and 2000 revolutions per minute (rpm) of not less than 6000W, 120/240 volts (V), 60 Hertz, single phase, ac. The alternator shall be rigidly coupled to the engine, permanently aligned and of single bearing design. The alternator shall include a 12V, direct current, 2 to 5 ampere (A), adjustable trickle charger which shall

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operate through the wiring harness connecting the starting system to the battery. The alternator shall have a brushless exciter and voltage regulated for ±3 percent maximum from no load through maximum rated load.

3.4.1 Power distribution equipment. All electrical components shall conform to applicable UL standards, including UL 486, UL 489, UL 498, and to NFPA 70. The installation of all electrical components and systems shall conform to NFPA 70. The power distribution equipment shall consist of the power distribution panel, ballasts for the metal halide lamps, switches, receptacles, cord connectors and plugs, portable power cords, interconnecting wiring, fitting, and accessories, installed and connected as required to provide a complete, operable system.

3.4.1.1 Power distribution panel. The power distribution panel shall be a circuit breaker panel mounted in a readily accessible location within the floodlight set housing. Sufficient circuit breakers shall be contained in the panel to provide independent control and circuit protection as defined in NFPA 70.

3.4.2.2 Circuit breakers. The circuit breakers shall conform to the requirements of NEMA AB 1, and UL 489. All circuit breakers shall be approved for use as switches and shall be marked as such. Circuit breakers shall be ventilated, temperature compensating, and sized to have trip rating as required for protection against overloads, short circuits, and grounding in accordance with NFPA 70.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5.1 Dissimilar metals. When dissimilar metals are used in contact with each other suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5.2 Environmental and storage conditions. All components of the floodlight set shall withstand, without damage or impairment of operation, various environmental conditions including the following:

a. Rated performance, without additional auxiliary equipment, in any ambient temperature from -25 degrees Fahrenheit (°F) to +125°F (-31.7° celsius (C) to +51.7°C) or when exposed to rainfall at a rate up to 3 inches per hourimpinging on the set at any angle from the vertical to 45° from the vertical.

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b. The set shall not be damaged by exposed storage at temperatures from -65°F to 155°F (-53.9°C to +68.3°C) at any possible relative humidity within this temperature range.
c. Manufacturers certification that the operating components furnished meet all the conditions stated in subparagraphs a. and b. will be acceptable.

3.6 Floodlight set components.

3.6.1 Generator set. The generator set shall provide adequate electrical power for operation of the floodlight set at rated maximum load during environmental and test conditions specified herein. The unit shall be mounted such as to minimize vibration. The set shall be furnished with an ac voltmeter and engine hour meter, as a minimum number of instruments.

3.6.1.1 Engine. The engine shall be a four-cycle, air-cooled diesel which develops, at between 1600 and 2000 rpm, the necessary horsepower to meet the requirements specified for the floodlight set.

3.6.1.1.1 Engine fuel. The diesel engine shall operate efficiently on DG-1, DG-2, DF-A, and DF-M as primary fuels, and be capable of operation on jet fuels JP-4, JP-5, and JP-8.

3.6.1.1.2 Fuel tank. The fuel tank shall be of sufficient capacity to permit full load operation for a minimum of 18 hours. The fuel tank shall be removable from the floodlight set to permit cleaning and shall contain a fuel shut-off valve at the outlet connection.

3.6.2 Wiring. Conductors for interconnecting wiring shall be composed of copper conductor and have a flame-retardant and oil-resistant covering conforming to NFPA 70. Extra flexible conductors shall be used where relative movement between parts may occur. Wire connectors shall be the crimp type where practicable and shall be attached in a manner to fully protect the insulation; all wire connectors shall be copper and conform to UL 486. Conductors shall be properly clamped in a neat manner and protected by grommets or other suitable protection from abrasion where the conductors pass through metal.

3.6.2.1 Ballasts. The ballasts for the four 1000 size metal halide lamps shall be the peak-lead type designed to operate with metal halide lamps and 120/240V input. The ballast shall be the enclosed type and shall be the constant wattage auto-transformer type, and shall be designed for operation at low temperature down to -25°F. The ballasts shall be mounted near the power distribution panel in a protected location inside the storage compartment. A double-throw switch shall be installed and connected in each ballast circuit to bypass the ballast, thereby enabling the use of incandescent lamps in the 1000W size luminaires, when required, and the switch will be protected to prevent accidental switching into the incandescent lamp position. The line side of the ballasts shall be connected to the circuit breakers in the distribution panel.

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3.6.2.2 Connectors, plugs, and receptacles. Cord receptacles, cord plugs, and outlet receptacles shall be of the grounding type and shall be in accordance with UL 498, W-C-596, NEMA WD 1 and WD 5, as applicable. The cord receptacles and cord plugs shall employ cord grip construction and shall be weatherproof. Cord plugs and cord receptacles shall have replaceable weatherproof covers made of black neoprene or acrylonnitrile-butadiene/ polyvinyl-chloride rubber and shall be so designed that they form a weatherproof seal when the plug is inserted into the mating receptacle either of another cord or of an outlet on the storage compartment exterior. All recptacles exposed to the weather shall be equipped with spring loaded cover plates.

3.6.2.2.1 Receptacles. Three grounding type 2 pole, 3 wire, 125V, 15A, general purpose, duplex, hospital grade, receptacles conforming to W-C-596/12 and NEMA WD 1 configuration 5-15R; one grounding type 2-pole, 3 wire, 125V, 15A locking type duplex receptacle conforming to W-C-596/99 and NEMA WD 5 configuration L5-15R; and two grounding type 3 pole, 4 wire, 480V, 20A, single locking type recptacles conforming to UL 498 and NEMA WD 5 configuration L16-20R shall be installed in the same vicinity of each other in an accessible, protected location on the exterior of the storage compartment. The locking type receptacles shall be used for the 1000W and 500W luminaries. The general purpose duplex receptacles shall be used for auxiliary equipment. Each outlet shall be marked for intended use and voltage. Each duplex receptacle shall be connected on only one circuit so that one circuit breaker protects one duplex receptacle load. Each 3 pole, 4 wire, 480V, 20A locking type receptacle shall be connected to the outputs of two ballasts which supply power to two 1000W luminaires. The input to each pair of ballasts shall be connected to a multi-wire 120/240V circuit with a common neutral so that two circuit breakers, one for each ungrounded conductor, protect the ballasts. Each of these receptacles shall supply power to two 1000W luminaires. The circuit breaker for each luminaire shall also function as the switch for turning on each floodlight lamp.

3.6.2.2.2 Portable power cords. Two or more portable power cords shall be provided for use as power extension cords for the 500W luminaires mounted on the portable mast. The cords shall total not less than 300 feet in length and shall be 3 conductor, No. 12 American Wire Gage (AWG) copper conductors. One end of each cord shall be terminated in a plug connector conforming to W-C-596/100 and NEMA WD 5, configuration L5-15P. The other end of each cord shall be terminated in a receptacle connector conforming to W-C-596/99 and NEMA WD 5, configuration L5-15R.

3.7 Storage compartment (housing). A durable metal housing shall be mounted on the trailer to house all permanently mounted power distribution equipment and provide for storage of all floodlights, electrical cords, etc. The housing shall contain hinged doors to allow easy access to stored components and allow access to all components necessary to utilize the floodlight set. The doors shall allow easy access to components requiring servicing. Doors shall be reinforced to provide adequate rigidity and have fasteners to hold then securely in the closed (locked) position. All latches

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and handles shall be proportioned to facilitate manipulation with heavy mittened hands and with a minimum of projection beyond the housing surface. Doors and hinged covers shall have a lock-open device to securely hold the door or cover in the open position with the generator set in operation. Drip channels or similar features shall be provided over vertically hinged doors and covers. Doors and covers shall be joined with the housing in a manner that will meet environmental requirements and reduce noise resulting from vibration. Means shall be provided for drainage of water entrapped within the storage compartment resulting from operation with doors or covers open. Hinges shall be of the continuous type. The storage compartment may be used for storage of the telescoping mast and the portable masts, provided they are adequately reinforced to provide structural support for extended off-road usage.

3.7.1 Compartment interior lighting. A small light shall be installed to illuminate the interior of the storage compartment. The light shall accommodate a 12V, direct current, incandescent lamp not larger than 40W. The light shall have a frosted dome lens, shall be controlled by a conveniently located toggle switch, and shall be protected by a fuse located at the attaching point to the power source.

3.7.2 Structural support. The compartment shall be reinforced as required to mount and support the lighting fixtures, extension cords and related gear when stowed in the travel positions. The storage compartment shall not deform, twist, or deflect during tests required in 4.6.5.

3.8 Trailer. The trailer chassis shall support the complete floodlight set, when subjected to the usage specified herein, without permanent distortion or damage to any part. The trailer shall be provided with a frame of all steel welded construction. The drawbar shall be of the A-frame type and be an extension of the main side frame members, and may be designed as a quick detachable type to assist the reduction of storage area required. The frame and drawbar shall have structural capacity and be reinforced or braced, where necessary, to produce adequate strength and rigidity to prevent both horizontal and vertical buckling or distortion. The floodlight set shall have a minimum road clearance of 8 inches. Springs shall be of a capacity to support the imposed load when measured at the ground. A lunette ring conforming to MS51336, with a 3-inch inside diameter eye fabricated from 1 5/8-inch diameter steel rod or tubing of not less than 3/16-inch wall thickness, and shall be installed and positioned for attachment to the towing vehicle in such a manner that the coupling height will be adjustable between 20 to 30 inches above the ground.

3.8.1 Landing gear. An adjustable length type landing leg, with a swivel caster wheel, of rugged construction and with provisions to hinge and lock in a vertical and horizontal position shall be provided and attached to the draw bar. The landing leg shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained and the forces resulting from impacts due to coupling, decoupling, transportation, and rough handling. The rearward, and side thrust of the landing leg. The landing leg, when in the vertical position, shall hold the trailer chassis relatively level and shall not obstruct towing in the horizontal position.

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3.8.2 Stabilization jacks. Adjustable swivel type jack stands shall be securely attached to extendable and retractable outriggers to level and stabilize the complete, free standing floodlight set on a 5° slope, when tested in accordance with 4.6.6. Means shall be provided for locking the jacks and outriggers in a retracted position which will not obstruct towing. When locking pins are used, they shall be of the captive type with chain. Level indicators shall be provided in accessible protected locations on the trailer chassis to facilitate leveling the floodlight set prior to starting operation. The jacks shall be equipped with swivel foot pads. The stabilization jacks shall provide sufficient support to prevent damage with full extended mast equipped with four floodlights during winds up to 30 miles per hour (mph) continuously from any side direction.

3.8.3 Wheels, rims, and tires. The trailer shall be mounted on two pneumatic-tired wheels with a minimum diameter of 14 inches. Wheels shall have ball or roller bearings and shall be demountable at the hub or rim. Rim and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be of the tubeless type, with actual load imposed on each tire, measured on each wheel at the ground, +10 percent for 50 mph highway travel. Tires shall be new, unblemished, of first quality, and shall contain anti-ozous compound. Recommended tire pressure shall be stenciled on the trailer near each tire location. A spare wheel and tire shall be furnished. The tires shall be furnished with a highway type tread design.

3.8.4 Fenders. The trailer shall be furnished with steel fenders providing protection from moisture or debris being splashed or thrown upon the outer surface of the storage compartment and trailer chassis. The fenders shall be constructed and installed in such a manner as to provide for abuse and usage as specified in 4.6.9.

3.8.5 Center of gravity. The center of gravity of the floodlight set shall be forward of the axle and with all components stowed for towing, the load on the lunette ring shall not exceed 200 lb or be not less than 100 lb.

3.8.6 Tracking ability. When towed on a hard surface highway in accordance with 4.6.10 any resulting forces produced on the towing vehicle shall not be of a magnitude or nature that would cause noticeable steering difficulties. When towed in a straight line on a level, smooth, paved surface, the floodlight set shall follow the towing vehicle without swerving over 3 inches to either side of the path of the towing vehicle (see 4.6.10).

3.8.7 Trailer lights, wiring, and reflectors. All trailer lights and reflectors shall conform to DoT Federal Motor Carrier Safety Regulations sections 393.14, 393.19, 393.20, 393.22 through 393.26(d) and the recommended practices and standards in the SAE handbook. A lighting cable connector conforming to SAE J560 shall be attached to the wiring system to permit connection to the towing vehicle. All wiring in the trailer lighting system shall conform to SAE J1292.

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3.8.8 Parking brake. Unless otherwise specified (6.2.1), a lever type handbrake shall be provided which will hold the floodlight set in a stationary position when tested in accordance with 4.6.11. The brake system shall be an automotive internal expanding shoe and drum type. Parking brake cables shall be made of stainless steel and firmly secured.

3.9 Telescoping tower and masts. A telescoping tower mounted on the floodlight set chassis, and two portable, collapsible telescopic masts shall be furnished. The telescoping tower shall be mechanical elevating and telescoping (see 3.9.1) or mechanical elevating and pneumatically telescoping (see 3.9.2) as specified. The tower assembly shall have provision for securely mounting four 1000W size luminaires in a horizontal configuration with a spacing between luminaires (center-to-center) of not less than 3 feet on the top of the tower when in operating position. The tower assembly shall have four grounding types, 2 pole, 3 wire, 480V, 20A, locking type, receptacles conforoming to UL 498 and NEMA WD 5, configuration L820R, mounted in four weatherproof boxes, one receptacle per box, near the upper end of the top section of the tower and on opposing sides of the tower. The boxes shall have weatherproof cover plates as specified in 3.6.3.5. Each box shall supply power to one 1000W luminaire. Two 4 conductor, No. 12 AWG copper conductor, heavy duty, flexible, permanently coiled, and retractable type power cords, one for each box, shall furnish power to the boxes. The supply end of each cord shall terminate in a weatherproof, grounding type, 3 pole, 4 wire, 480V, 20A, locking type plug conforming to UL 498 and NEMA WD 5, configuration L16-20P. The retractable power cords shall have guide tubes, one on each side of the tower, which encase the power cord in the completely retracted position, service the guide, and prevent entanglement of the power cords when the telescoping tower is in the extended position. Not less than 6 feet of the cord slack, at the plug end of each cord, shall be provided to connect the cords to the 3 pole, 4 wire, 480V, 20A power receptacles on the exterior of of the storage compartment. Provision shall be made for securing the cord slack when the floodlight set is in transit.

3.9.1 Mechanical telescoping tower. Unless otherwise specified (see 6.2.1), the telescoping tower shall be a fabricated assembly that, when in the upright (operating) position, is continuously adjustable from a fully lowered position of not more than 13 feet to a height of not less than 30 feet above ground. The lower portion of the tower shall be enclosed in a manner to protect operating personnel from injury while the tower is being extended or retracted. The tower assembly shall be hinged or pivoted in such a manner that, with the tower assembly in the fully retracted position, the assembly may be lowered into a horizontal (transport) position. The floodlight set shall have a positive locking mechanism to secure the tower assembly for all specified operating conditions and secure the tower assembly to the trailer-storage compartment assembly for all specified transport conditions. The floodlight set shall have a manual tower operating mechanism such that one man can raise the tower assembly from the horizontal (transport) vertical position and extend the tower to the maximum height in not more than 5 minutes. The tower assembly shall have a tensioning mechanism to assure that

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the tower assembly is completely under control of the operating mechanism during extending or retracting operations. The tower assembly shall have a positive means, independent of the extending-retracting mechanism, to lock the tower assembly sections in the fully retracted position, the fully extended position, and not less than three positions; positions approximately equally spaced between the fully extended and fully retracted positions. The locking mechanism may be either automatic or controlled from the tower operating position. The tower assembly shall have structural strength to withstand 30 mph winds from any direction, without damage, when fully extended with the luminaires mounted thereon, and shall have structural strength to withstand 60 mph winds from any direction without damage when extended to the midpoint between minimum and maximum operating positions and with the luminaires mounted thereon.

3.9.2 Pneumatically telescoping tower. When specified (see 6.2.1), the tower shall be a telescoping member consisting of multiple sections of heavy duty aluminum tubing; internally honed and externally anodized, clear sealed. The tower shall have tubular overlapping sections with two external full length keyways to prevent horizontal deflection. It shall contain internal and external bumpers to minimize shock during raising and lowering. There shall be no metal to metal contact. Raising and lowering of the tower shall be accomplished pneumatically by compressed air. Maximum required air pressure shall be 20 pound-force per square inch and the tower must raise to maximum height in no more than 3 minutes. The tower must be adjustable from the fully lowered position of 8 $\pm$1/2 foot to a height of 30 $\pm$2 feet above the ground and any position in between. The tower must be able to be rotated 340° (without lowering the mast) by an operator standing on the ground and shall contain a floodlight mounting bar at the top which may be rotated around its longitudinal axis to point the floodlights straight up, straight ahead, and downward. Four 1000W floodlights shall be mounted to the bar in a box like configuration. The tower must be designed to be removable from its vertical (in use) position for placing it in its stowed position for floodlight set transportation. The tower weight shall be such that one man can lift the tower to perform this transfer. The tower shall be designed so that it may be serviced and overhauled by maintenance personnel at field level. When the pneumatically telescoping tower is specified, the floodlight set shall include an air compressor (see 3.9.2.2).

3.9.2.1 Tower settling. With the floodlight set fully configured for use, the telescoping tower raised to maximum height and the engine turned off, the tower shall not settle (lower) more than 1/4 inch per hour over a 12-hour period.

3.9.2.2 Tower air compressor. The floodlight set shall contain an electrically driven air compressor of sufficient capacity to operate the telescoping tower. Means shall be provided to prevent over pressurization of the telescoping tower and, in the event of failure or malfunction of any component, to prevent the tower from descending at a rate that would cause damage to any component or create a safety hazard.

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3.9.3 Tower electrical cord. The electrical cord supplying power to the telescoping tower floodlights shall be restrained along the tower mast to prevent wind whip. It shall be connected to the tower floodlight mounting bar in a manner to prevent cord damage and separation from the bar.

3.9.4 Portable masts. Two portable mast assemblies with tripod base, shall be provided for mounting two 500W size luminaires each. The mast assembly shall consist of a manually adjustable telescoping tubular mast with a tubular crossarm on which to mount the luminaires and a collapsible tripod base. The height of the mast shall be manually adjustable in increments of not more than 6 inches from a minimum of 8 feet to a maximum of 12 feet, and shall have a locking mechanism to securely lock the mast at any of the selected positions. The tripod base, when in the operating position, shall securely support the mast in a vertical position at maximum height with the luminaires mounted on the crossarm and shall provide sufficient stability at this position to withstand 10 mph winds from any direction. The mast, when fully retracted, with two luminaires mounted on the crossarm and the base anchored shall have sufficient structural strength to withstand 30 mph winds from any direction. The portable mast assembly shall be readily collapsible for transport storage in storage compartment (see 3.7). The collapsed mast assembly shall require minimum storage space with an overall length of not more than 6 feet.

3.10 Floodlights. The floodlight set shall have four 1000W size floodlights and four 500W size floodlights. Each floodlight shall consist essentially of a cast, pressed or spum aluminum, weatherproof housing with cooling fins, if required, and a hinged door, with a heat-resistant clear glass lens, and weather-resistant seal. The door shall be secured to the housing with not less than two securing devices. The securing devices shall be captive with either the housing or door. Provisions shall be made to release the securing devices and remove the door without tools. The securing devices shall maintain the weather-resistant seal under the wide temperature changes, adverse weather, shock, and vibrational stresses encountered under rough service conditions.

3.10.1 1000-watt floodlights. The floodlight mounting arrangement shall be such, that the four lights attached to the tower may be individually rotated through an arc of not less than 350° in the horizontal plane, and an arc of not less than 135° in the vertical plane. The housing shall have a suitable carrying handle. The floodlight shall have a lateral and vertical beam spread of not less than 120° ±10°. The clear glass lens shall produce a controlled circular beam spread. The floodlight shall be provided with a keyless mogul base lampholder to accommodate a metal halide lamp or an incandescent lamp. The floodlight shall have a shock mount lamp support, and a heat-tempered clear glass lens. Adequate provisions shall be available for front or rear relamping. Each floodlight shall have a flexible power supply cord attached with a watertight cord grip at the floodlight end and a grounding type 2 pole, 3 wire, 480V, 20A, locking type cord plug with weatherproof cover on the supply end. The plug shall conform to UL 498 and

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NEMA WD 5 configuration, L820P. The cord shall be 3 conductor, No. 14 AWG copper, and as specified in 3.6.3.5. The green grounding conductor at the floodlight end shall be connected to the floodlight housing. Metal halide lamps which produce white light shall be used. High pressure sodium or low pressure sodium lamps shall not be used. The initial light output for 1000W lamps shall be not less than 200,000 lumens.

3.10.2 500 watt floodlight. The four 500W floodlights shall be designed to be mounted on the portable mast specified in 3.9.4. A mounting bracket, adequate to support the floodlight on the portable mast shall be provided. With the portable mast resting in a horizontal plane, vertical adjustment, without tools, of the floodlight through a range of not less than 60° above to 30° below the horizontal shall be provided. There shall be a means to securely lock the floodlight in the mounting bracket at any selected position. Horizontal adjustment, without tools of the mounting bracket with respect to the portable mast through a full 360° shall be provided. There shall be a means to securely lock the mounting bracket to the portable mast at any selected position. The floodlight shall be designed to use 120V clear, T-3 rough service, quartz, tungston Halogen lamps with recessed single contact base. The initial light output for 500W lamps shall be not less than 10,900 lumens. Each shall have a flexible 25 foot long power supply cord conforming to UL 62, attached with a watertight cord grip at the floodlight end and a grounding type, 2 pole, 3 wire, 125V, 15A, locking type cord plug with weatherproof cover on the supply end. The plug shall conform to W-C-596/100 and NEMA WD 5, configuration L5-15P. The cord shall be 3 conductor, No. 14 AWG copper. The green grounding conductor at the floodlight end shall be connected to the floodlight housing.

3.10.3 Lamps. Lamps which produce white light shall be furnished installed in all floodlights. All lamps shall be designed to operate in a horizontal position. Provision shall be made to stow one set of spare lamps in the storage compartment.

3.11 Air transportability. When specified (see 6.2.1), the floodlight set shall be designed to be air transportable in accordance with MIL-A-8421. Tiedown points, fore, aft, and sidewards shall be marked on the set and shall include the g loads for which each point is designed.

3.12 Lifting and tiedown attachments. When specified (see 6.2.1), the floodlight set shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a plate conforming to the vehicle shipping data plate described in MIL-STD-209, except tiedown attachments may be identified by stenciling or other suitable marking on the floodlight set. Tiedown marking shall clearly indicate that the attachments are intended for tiedown of the floodlight set on the carrier when shipped.

3.13 Lubrication. Unless otherwise specified in section 5, lubrication shall be in accordance with the equipment manufacturer's recommendations.

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3.14 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.15 Instruction plates. The floodlight shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.15.1 Identification plate. An identification plate will be furnished by the contracting officer for each floodlight. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each floodlight in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.15.2 Operating instructions. Plastic encapsulated operating instructions shall be mounted in a convenient place inside the storage compartment. The operating instructions shall be complete and brief.

3.15.3 Schemetic and wiring diagrams. A plastic encapsulated schematic wiring diagram shall be mounted adjacent to the operating instructions. The schematic diagram shall show the complete operation and functional sequence of the circuit for analysis and maintenance of the floodlight set. The wiring diagram shall show the physical location of all wiring interconnections in the same relationship in which they are installed. Identification marking of control devices and connections shall be shown in both diagrams and shall coincide with markings on all items being identified.

3.15.4 Technical manuals. Technical manuals shall be furnished as specified in the contract (see 6.3).

3.16 Rustproofing. When specified (see 6.2.1), rustproofing shall be provided in accordance with MIL-STD-1223.

3.17 Electromagnetic interference. The floodlight set, fully extended and operating, shall meet the requirements of MIL-STD-461, class C-3 UM05 for conductive interference and SAE J551 for radiated interference.

3.18 Servicing and adjusting. Unless otherwise specified in section 5 and prior to acceptance of the floodlight sets by the government, the contractor shall service and adjust the floodlight sets for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

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1. Inflation of all tires.
2. Adjustment of brakes.
3. Proper functioning of all lighting and electrical systems.

3.19 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2.1). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. Color of the finish coat shall be Green, No. 14064, or Yellow, No. 13538, conforming to FED-STD-595 'Colors'. The end item, allied equipment, and attachments shall be the same color.

3.20 Workmanship.

3.20.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.20.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.20.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.20.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one floodlight set (see 3.2, 6.2.1, and 6.4). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.4 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.6.14, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.3.

4.5 Examination. Each floodlight shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Unless otherwise specified (see 6.2.1), the first article floodlight set shall be tested as specified herein. Failure of the floodlight set to meet requirements of the tests herein, even though not performing a test directly related to the specific requirement, shall constitute cause for rejection. The contractor is responsible for providing certification or evidence of conformance to all the requirements specified.

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4.6.1 Engine and components test. The engine and components shall be inspected for conformance to requirements of 3.6.1.1 through 3.6.1.1.2.

4.6.2 Alternator test. The alternator shall be inspected for conformance to 3.4.

4.6.3 Power distribution equipment test. All electrical components shall be inspected for conformance to 3.6.2 through 3.6.2.2.2. After verification of conformance the floodlight set shall be assembled with all lights connected to the proper outlets. The floodlight set shall be cycled 250 times in 25 separate 10-cycle runs without failure or malfunctioning of any component. A 10-cycle run is as follows:

a. Start engine and turn on the four tower floodlights and the four portable mast floodlights.
b. When equipped with pneumatically telescoping tower, the tower shall be elevated to maximum height and lowered to minimum height, 10 times, each cycle.
c. When equipped with mechanical telescoping tower the tower shall be raised to maximum height and lowered to minimum height, one time, each cycle.
d. Turn off lights and shut down engine.
e. Leave engine shut down for 5 minutes.

Note: A minimum of 10 cycles each with engine operating on DF-2 and JP-4 fuels shall be performed during the test operation of 4.6.3.

4.6.4 Power cords test. The extension power cords shall be inspected for conformance with 3.6.2.2.

4.6.5 Storage compartment test. The compartment shall be inspected for conformance with requirements of 3.7.2. The floodlights and all specified and required gear shall be placed in the designated areas and inspected for adequate fit and support. The interior lamp shall be tested for function. Upon completion of the road tests (see 4.6.10) the compartment shall be inspected for cracking, deforming, deflection, twisting and malfunction of the compartment doors and latches. Any sign of distortion or damage shall be cause for rejection.

4.6.6 Stabilization jack test. With the fully elevated floodlight set uncoupled and engine operating and the outriggers extended, the jacks shall prevent any movement of the floodlight set. The fully erected and uncoupled floodlight set shall be subjected to a stability test in accordance with the requirements of 3.8.2.

4.6.7 Telescoping tower tests. With the floodlight set free standing on a relative smooth unpaved area having a slope of approximately 5°, the outriggers extended, and the set leveled, the mechanism shall be actuated and the tower raised to its extreme extended position. While extended, a force of 40 lb, to simulate 30 mph wind conditions shall be applied successively to the

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tower at not less than four locations spaced equidistant around its periphery, perpendicular to the tower and at crossarm height, to determine conformance to rigidity and construction requirements. The telescoping tower shall be lowered from the fully extended position to its mid-point and a force of 75 lb applied at four successive locations around the tower periphery to simulate 60 mph wind conditions. Similar checks shall be made to determine stability of the tower at the 60 mph wind loading.

4.6.7.1 Pneumatic tower simulated failure test. With the floodlight set fully erected for use and the telescoping tower raised to maximum height, a failure of the compressor air supply line shall be simulated. The floodlight set shall meet the requirements of 3.9.2. All air control valves between the compressor and the tower shall be fully open for this test.

4.6.8 Center of gravity test. With all components of the floodlight set stowed for normal transporting, the load on the lunette ring shall be measured to determine conformance to 3.8.5.

4.6.9 Fender test. The fenders shall be subjected to a static test of a 150 lb load on each fender for a continuous period of 8 hours or a dynamic load of 75 lb sand bag or similar item attached to the top surface of the fender during the road tests. Any sign of deformation or support failure shall be cause for rejection.

4.6.10 Road test. All components shall be stowed or attached in the normal manner for transporting, with fender weights attached (see 4.6.9), and the floodlight set coupled to a suitable towing vehicle. The floodlight set shall be towed a distance of not less than 100 miles over improved roads at speeds up to and including 50 mph (see 6.5.1), tracking ability shall be tested for conformance to 3.8.6. In addition, the floodlight set shall be towed a distance of not less than 30 miles over unimproved roads at speeds up to and including 15 mph (see 6.5.2).

4.6.11 Brake holding capacity test. With the complete floodlight set parked on a 15°, dry, hard-surfaced incline and with the wheels facing down the slope, the handbrake shall be set and the trailer uncoupled from the towing vehicle and rested on the landing leg to determine conformance to 3.8.8.

4.6.12 Portable mast stability test. The portable telescoping mast shall be fully extended and be subjected to the stability requirements specified in 3.9.4.

4.6.13 Electromagnetic interference test. The floodlight set shall be tested to show compliance with the requirements of 3.17.

4.6.14 Quality conformance test. Each production floodlight set shall be given an operational test. All components, and devices shall be operated to determine that they are functioning properly. The electrical system shall be tested for grounds, open circuits, and short circuits. Any set which fails to pass any of the tests shall be rejected.

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4.6.15 UL certification. Acceptable evidence of meeting the applicable UL standards (see 3.6.2) shall be the UL listing mark affixed to the item certified, or a certified test report from a nationally recognized independent testing laboratory, acceptable to the contracting officer, stating that the item certified has been tested and conforms to the applicable UL standards. When a certified test report is submitted, it shall be furnished as specified in the contract (see 6.3)

4.7 Packaging inspection. The preservation, packing, and marking of the floodlight set shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1.

5.1.1.5 Generator set. The generator set shall be preserved method I in accordance with the level A requirements of MIL-G-28554 for generator set uncrated.

5.1.1.6 Floodlights, tower, and masts. The floodlights shall be stored in the storage compartment and secured to prevent damage during transit. The telescoping tower shall be retracted to the shipping position and secured. Portable masts shall be collapsed and secured in the storage compartment.

5.1.1.7 Power cords and spare lamps. The power cord and any spare lamps shall be secured in the storage compartment to prevent damage during transit.

5.1.1.8 Jacks. The shaft of the jack shall be extended and preserved with type P-11 preservative. The shaft shall then be fully retracted.

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5.1.1.9 Trailer. The trailer shall be preserved in accordance with the level A (mobile) requirements of MIL-STD-281.

5.1.1.10 Lubrication. Lubricate all grease fittings with type P-11 preservation. Wheel bearings requiring a NLGI No. 2 grade grease shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.11 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 lb above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

5.1.1.12 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.13 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.14 Consolidation. Loose components, tools, and publications for each floodlight set shall be consolidated in containers conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186.

5.1.2 Level C. The floodlight set shall be preserved in accordance with the contractor's standard practice in a manner to prevent corrosion, deterioration, and damage. The floodlight set shall be lubricated and serviced for operation as required in the operator's manual and specified in section 3.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Crated floodlight sets. Unless otherwise specified (see 5.2.1.2 and 6.2.1), each floodlight set shall be packed in a crate conforming to MIL-C-104, type II, class 1 or 2, style A. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the appendix to MIL-C-104. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B. The floodlight set shall not be crated resting on its supporting legs (landing gear).

5.2.1.2 Mobile (uncrated floodlight sets. When specified (see 6.2.1), each floodlight set shall be prepared for mobile shipment in accordance with applicable carrier rules and regulations. Consolidation containers shall be secured on the floodlight set in a manner not to increase cubage or interfere with towing or lifting the floodlight set with slings. Do not ship the floodlight set resting on its support legs (landing gear).

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5.2.2 Level B.

5.2.2.1 Crated floodlight sets. Unless otherwise specified (see 5.2.2.2 and 6.2.1), each floodlight set shall be packed in a crate conforming to MIL-C-3774, type II, style A. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the appendix to MIL-C-3774. The floodlight set shall not be crated resting on its support legs (landing gear).

5.2.2.2 Mobile (uncrated) floodlight sets. When specified (see 6.2.1), each floodlight set shall be prepared for mobile shipment as specified for level A.

5.2.2.3 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations. Do not ship the floodlight set resting on its support legs (landing gear).

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The floodlight set is intended for use at shore establishments and for military field application under all kinds of climatic conditions. It is a completely self-contained unit, suitable for use as a source of electrical floodlighting in remote locations, with electrical power as an auxiliary feature.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is not required (see 3.2, 4.3, and 6.4), when number of units, or types of tests vary (see 4.6).
c. When a parking brake is not required (see 3.8.8).
d. When mechanical telescoping tower required (see 3.9.1).
e. When pneumatically telescoping tower required (see 3.9.2).
f. When air transportability is required (see 3.11).
g. When lifting and tiedown is required (see 3.12).
h. When fungus resistance treatment required (see 3.14).
i. When rustproofing required (see 3.16).
j. When color requirement other than No. 14064 green (see 3.19).
k. Level of preservation and level of packaging required (see 5.1 and 5.2).
l. When mobile (uncrated) shipments are required (see 5.2.1.2 through 5.2.2.2).

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6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved DD Form 1664, Data Item Description (DID), and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirements title	Applicable DID No.
3.17.4	Manual, Technical, Modified Commercial	DI-M-24006E
4.5 and 4.6.15	Certificate of Compliance	DI-E-2121

(DIDs related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, Acquisition Management Systems and Data Requirements Control List. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.4 First article. When a first article inspection is required, the floodlight will be tested and approved under the appropriate provisions of supplement 27.410-5 of the FAR and should be a first production item as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Navy - YD

Review Activity:

DLA - GS

User Activity:

Navy - MC

Preparing Activity:

Navy - YD

(Project 6230-N304)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

6230-N304

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-29161C	FLOODLIGHT SET, ELECTRIC, DIESEL-ENGINE-DRIVEN, TRAILER MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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